

Data Website Project

Financial Data Analytics

Overview

You are an analyst tasked with building a public website around a data set you choose. The site has two pages. The first page is a report that the reader scrolls through, with your findings, the numbers behind them, and a chart for each. The second page is a dashboard that loads the data in the browser and lets the reader filter it, switch between charts, and see the numbers change.

Projects are 50% of your course grade, and the project grades are averaged.

You will build the site in a public GitHub repository and publish it with GitHub Pages. You will turn in one text file (.txt or .md) with four lines: your name, your student ID, the repository URL, and the live site URL.

The due date is posted on Blackboard. No late submissions are allowed.

The data

Choose any data set that interests you. It can be about anything: sports, music, movies, video games, weather, traffic, elections, a business, a market. The HMDA mortgage data we use in class is not allowed. The data set must meet the following requirements:

1. The data must be panel data. The file has a time column (a year, a season, a month, a week, a date) and a group column (a company, a team, a player, a country, a city, a store, a product), where the same groups appear period after period. One row is one group in one period, for example one company in one year, one player in one season, one team in one game, one store in one week. Alternatively, one row is one event, such as a sale, a game, or an application, with the date it happened and the group it belongs to.
2. The time column has at least five different periods (five years, five seasons, five months).
3. The group column has at least ten different values.
4. The data set has at least 50,000 rows and at least eight columns. Among the columns are at least two categorical variables the reader can filter on and at least two numeric variables you can total, average, or rank.

If you find a data set you really want to use and it does not meet one of the requirements, email me before you start so I can approve it.

The report page

The report is `index.html`, the page that opens at the site URL. It holds the following:

1. A title, your name, and a one-paragraph summary of what the data covers and what you found.
2. A block of at least four headline numbers, each with a label that says what it is.
3. At least eight sections. Each section has a heading that states the finding, one or two paragraphs that explain the finding and give the numbers behind it, and a chart of those numbers.
4. A closing section that explains your data set: where the data comes from, what one row is, any rows you drop and why, and how you compute every rate, ratio, or average you report.

The dashboard page

The dashboard is `dashboard.html`, linked from the report page. It loads the data, does calculations in the browser, and holds the following:

1. Filters for at least four variables, among them the time column and the group column.
2. At least four summary numbers that change with the filters.
3. At least four charts that change with the filters, with a switch that changes the measure shown (for example a count, a total, a median, a rate) and a switch that changes the variable the chart is broken down by.
4. A table that shows the numbers behind the current view.
5. A button that resets every filter.

The site as a whole

- Both pages share one navigation bar and one set of fonts and colors. The look is up to you. Have fun with it.
- I recommend building the site with plain HTML, CSS, and JavaScript, but you are free to use anything else.

The repository

The project lives in its own folder under your home folder, for example `~/stock-analysis`, and that folder is the repository. Do not build it inside the `~/fda-python` course folder.

- Public, with GitHub Pages enabled from the main branch.
- A `README.md` that lists every file and what it does, and says where the data came from.
- The data files, any scripts, and the site files.
- A commit history that shows the work as it was done, not one commit holding the finished site.

Using AI tools

You are expected to build the site with Claude Code or Codex. The data set, the questions you ask of it, the findings you report, the numbers, and the design are yours, and you are responsible for every number and every file in the repository.

Grading

The project is graded out of 100 points.

Component	Points
Report: findings, summaries, and charts	25
Report: numbers correct and reproducible from the data	20
Dashboard: filters, switches, and charts that work	25
Dashboard: numbers agree with the report and the data	10
Design and personalization	10
Repository: README, files, commit history	10

A page that works but whose numbers are wrong, or a page whose numbers are right but that does not work at the site URL, earns zero credit for that component.

Academic integrity

The site is your own work. You may discuss data sets, tools, and publishing problems with classmates, but do not share code or files.